

london_EDA

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2025-04-10

Loading & Merging Data

London census 2021 data merged with postal_codes and house_prices data.

Brief Intro to postal_codes & house_prices

```
head(postal_codes)
```

```
## # A tibble: 6 x 60
##   Postcode `In Use?` Latitude Longitude Easting Northing `Grid Ref` County
##   <chr>    <chr>      <dbl>    <dbl>    <dbl>    <dbl> <chr>    <chr>
## 1 BR1 1AA Yes        51.4     0.0154  540291   168873 TQ402688 Greater Lon~
## 2 BR1 1AB Yes        51.4     0.0152  540262   169405 TQ402694 Greater Lon~
## 3 BR1 1AD No         51.4     0.0167  540386   168710 TQ403687 Greater Lon~
## 4 BR1 1AE Yes        51.4     0.0142  540197   169204 TQ401692 Greater Lon~
## 5 BR1 1AF Yes        51.4     0.0149  540259   168855 TQ402688 Greater Lon~
## 6 BR1 1AG Yes        51.4     0.0149  540259   168855 TQ402688 Greater Lon~
## # i 52 more variables: District <chr>, Ward <chr>, `District Code` <chr>,
## #   `Ward Code` <chr>, Country <chr>, `County Code` <chr>, Introduced <date>,
## #   Terminated <date>, Parish <chr>, `National Park` <lgl>, Population <dbl>,
## #   Households <dbl>, `Built up area` <chr>,
## #   `Lower layer super output area` <chr>, `Rural/urban` <chr>, Region <chr>,
## #   Altitude <dbl>, `London zone` <dbl>, `LSOA Code` <chr>,
## #   `Local authority` <lgl>, `MSOA Code` <chr>, ...
```

```
colnames(postal_codes)
```

```
## [1] "Postcode"
## [3] "Latitude"
## [5] "Easting"
## [7] "Grid Ref"
## [9] "District"
## [11] "District Code"
## [13] "Country"
## [15] "Introduced"
## [17] "Parish"
## [19] "Population"
## [21] "Built up area"
## [23] "Rural/urban"
## [25] "Altitude"
## [27] "LSOA Code"
## [29] "MSOA Code"
## [31] "Parish Code"
## [33] "Index of Multiple Deprivation"
## [35] "User Type"

## "In Use?"
## "Longitude"
## "Northing"
## "County"
## "Ward"
## "Ward Code"
## "County Code"
## "Terminated"
## "National Park"
## "Households"
## "Lower layer super output area"
## "Region"
## "London zone"
## "Local authority"
## "Middle layer super output area"
## "Census output area"
## "Quality"
## "Last updated"
```

```
## [37] "Nearest station"          "Distance to station"
## [39] "Postcode area"            "Postcode district"
## [41] "Police force"             "Water company"
## [43] "Plus Code"                "Average Income"
## [45] "Sewage Company"           "Travel To Work Area"
## [47] "ITL level 2"              "ITL level 3"
## [49] "UPRNs"                    "Distance to sea"
## [51] "LSOA21 Code"              "Lower layer super output area 2021"
## [53] "MSOA21 Code"              "Middle layer super output area 2021"
## [55] "Census output area 2021"  "IMD decile"
## [57] "Constituency Code 2024"   "Constituency Name 2024"
## [59] "Property Type"            "Roads"
```

```
head(house_prices)
```

```
## # A tibble: 6 x 28
##   fullAddress      postcode country outcode latitude longitude bathrooms bedrooms
##   <chr>            <chr>    <chr>  <chr>      <dbl>    <dbl>      <dbl>    <dbl>
## 1 Flat 9, 35 Fur~ EC4A 1JQ England EC4A      51.5   -0.110         1         1
## 2 Flat 6, 35 Fur~ EC4A 1JQ England EC4A      51.5   -0.110        NA         NA
## 3 Flat 35, Octav~ SW1P 2TA England SW1P      51.5   -0.132         2         2
## 4 24 Chester Cou~ SE5 7HS  England SE5      51.5   -0.0922        1         1
## 5 18 Alexandra G~ N10 3RL  England N10      51.6   -0.140         1         4
## 6 17 Collins Str~ SE3 OUG  England SE3      51.5    0.00697        1         2
## # i 20 more variables: floorAreaSqM <dbl>, livingRooms <dbl>, tenure <chr>,
## #   propertyType <chr>, currentEnergyRating <chr>,
## #   rentEstimate_lowerPrice <dbl>, rentEstimate_currentPrice <dbl>,
## #   rentEstimate_upperPrice <dbl>, saleEstimate_lowerPrice <dbl>,
## #   saleEstimate_currentPrice <dbl>, saleEstimate_upperPrice <dbl>,
## #   saleEstimate_confidenceLevel <chr>, saleEstimate_ingestedAt <dtm>,
## #   saleEstimate_valueChange.numericChange <dbl>, ...
```

```
colnames(house_prices)
```

```
## [1] "fullAddress"
## [2] "postcode"
## [3] "country"
## [4] "outcode"
## [5] "latitude"
## [6] "longitude"
## [7] "bathrooms"
## [8] "bedrooms"
## [9] "floorAreaSqM"
## [10] "livingRooms"
## [11] "tenure"
## [12] "propertyType"
## [13] "currentEnergyRating"
## [14] "rentEstimate_lowerPrice"
## [15] "rentEstimate_currentPrice"
## [16] "rentEstimate_upperPrice"
## [17] "saleEstimate_lowerPrice"
## [18] "saleEstimate_currentPrice"
## [19] "saleEstimate_upperPrice"
## [20] "saleEstimate_confidenceLevel"
## [21] "saleEstimate_ingestedAt"
```

```
## [22] "saleEstimate_valueChange.numericChange"
## [23] "saleEstimate_valueChange.percentageChange"
## [24] "saleEstimate_valueChange.saleDate"
## [25] "history_date"
## [26] "history_price"
## [27] "history_percentageChange"
## [28] "history_numericChange"
```

YEAR 2021 DATA ALL MERGED.

```
head(london_2021_supra)
```

```
## # A tibble: 6 x 344
##   fullAddress      postcode country outcode latitude longitude bathrooms bedrooms
##   <chr>           <chr>    <chr>  <chr>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Flat 9, 35 Fur~ EC4A 1JQ England EC4A      51.5   -0.110         1         1
## 2 Flat 6, 35 Fur~ EC4A 1JQ England EC4A      51.5   -0.110        NA         NA
## 3 Flat 35, Octav~ SW1P 2TA England SW1P      51.5   -0.132         2         2
## 4 24 Chester Cou~ SE5 7HS  England SE5       51.5   -0.0922        1         1
## 5 18 Alexandra G~ N10 3RL  England N10       51.6   -0.140         1         4
## 6 17 Collins Str~ SE3 OUG  England SE3       51.5    0.00697        1         2
## # i 336 more variables: floorAreaSqM <dbl>, livingRooms <dbl>, tenure <chr>,
## #   propertyType <chr>, currentEnergyRating <chr>,
## #   rentEstimate_lowerPrice <dbl>, rentEstimate_currentPrice <dbl>,
## #   rentEstimate_upperPrice <dbl>, saleEstimate_lowerPrice <dbl>,
## #   saleEstimate_currentPrice <dbl>, saleEstimate_upperPrice <dbl>,
## #   saleEstimate_confidenceLevel <chr>, saleEstimate_ingestedAt <dtm>,
## #   saleEstimate_valueChange.numericChange <dbl>, ...
```

```
colnames(london_2021_supra)
```

```
##   [1] "fullAddress"
##   [2] "postcode"
##   [3] "country"
##   [4] "outcode"
##   [5] "latitude"
##   [6] "longitude"
##   [7] "bathrooms"
##   [8] "bedrooms"
##   [9] "floorAreaSqM"
##  [10] "livingRooms"
##  [11] "tenure"
##  [12] "propertyType"
##  [13] "currentEnergyRating"
##  [14] "rentEstimate_lowerPrice"
##  [15] "rentEstimate_currentPrice"
##  [16] "rentEstimate_upperPrice"
##  [17] "saleEstimate_lowerPrice"
##  [18] "saleEstimate_currentPrice"
##  [19] "saleEstimate_upperPrice"
##  [20] "saleEstimate_confidenceLevel"
##  [21] "saleEstimate_ingestedAt"
##  [22] "saleEstimate_valueChange.numericChange"
##  [23] "saleEstimate_valueChange.percentageChange"
##  [24] "saleEstimate_valueChange.saleDate"
##  [25] "history_date"
```

```

## [26] "history_price"
## [27] "history_percentageChange"
## [28] "history_numericChange"
## [29] "In Use?"
## [30] "Latitude"
## [31] "Longitude"
## [32] "Easting"
## [33] "Northing"
## [34] "Grid Ref"
## [35] "County"
## [36] "District"
## [37] "Ward"
## [38] "District Code"
## [39] "Ward Code"
## [40] "Country"
## [41] "County Code"
## [42] "Introduced"
## [43] "Terminated"
## [44] "Parish"
## [45] "National Park"
## [46] "Population"
## [47] "Households"
## [48] "Built up area"
## [49] "Lower layer super output area"
## [50] "Rural/urban"
## [51] "Region"
## [52] "Altitude"
## [53] "London zone"
## [54] "LSOA Code"
## [55] "Local authority"
## [56] "MSOA Code"
## [57] "Middle layer super output area"
## [58] "Parish Code"
## [59] "Census output area"
## [60] "Index of Multiple Deprivation"
## [61] "Quality"
## [62] "User Type"
## [63] "Last updated"
## [64] "Nearest station"
## [65] "Distance to station"
## [66] "Postcode area"
## [67] "Postcode district"
## [68] "Police force"
## [69] "Water company"
## [70] "Plus Code"
## [71] "Average Income"
## [72] "Sewage Company"
## [73] "Travel To Work Area"
## [74] "ITL level 2"
## [75] "ITL level 3"
## [76] "UPRNs"
## [77] "Distance to sea"
## [78] "LSOA21 Code"
## [79] "Lower layer super output area 2021"

```

```

## [80] "MSOA21 Code"
## [81] "Middle layer super output area 2021"
## [82] "Census output area 2021"
## [83] "IMD decile"
## [84] "Constituency Code 2024"
## [85] "Constituency Name 2024"
## [86] "Property Type"
## [87] "Roads"
## [88] "local authority code"
## [89] "local authority name"
## [90] "Distance traveled to work_All usual residents aged 16+ in employment"
## [91] "Distance traveled to work_Less than 2km"
## [92] "Distance traveled to work_2km to less than 5km"
## [93] "Distance traveled to work_5km to less than 10km"
## [94] "Distance traveled to work_10km to less than 20km"
## [95] "Distance traveled to work_20km to less than 30km"
## [96] "Distance traveled to work_30km to less than 40km"
## [97] "Distance traveled to work_40km to less than 60km"
## [98] "Distance traveled to work_60km and over"
## [99] "Distance traveled to work_Works mainly from home"
## [100] "Distance traveled to work_Other"
## [101] "Economic Activity_All usual residents aged 16 or over"
## [102] "Economic Activity_Economically active: Employee: Full-time"
## [103] "Economic Activity_Economically active: Employee: Part-time"
## [104] "Economic Activity_Economically active: Full-time student"
## [105] "Economic Activity_Economically active: Self-employed with employees: Full-time"
## [106] "Economic Activity_Economically active: Self-employed with employees: Part-time"
## [107] "Economic Activity_Economically active: Self-employed without employees: Full-time"
## [108] "Economic Activity_Economically active: Self-employed without employees: Part-time"
## [109] "Economic Activity_Economically active: Unemployed"
## [110] "Economic Activity_Economically inactive: Long-term sick or disabled"
## [111] "Economic Activity_Economically inactive: Looking after home or family"
## [112] "Economic Activity_Economically inactive: Other"
## [113] "Economic Activity_Economically inactive: Retired"
## [114] "Economic Activity_Economically inactive: Full-time students"
## [115] "hours worked_All usual residents aged 16+ in employment"
## [116] "hours worked_15 hours or less"
## [117] "hours worked_16 to 30 hours"
## [118] "hours worked_31 to 48 hours"
## [119] "hours worked_49 or more hours"
## [120] "Method of travel to work_All usual residents aged 16+ in work"
## [121] "Method of travel to work_Work mainly at or from home"
## [122] "Method of travel to work_Underground; metro; light rail; tram"
## [123] "Method of travel to work_Train"
## [124] "Method of travel to work_Bus; minibus or coach"
## [125] "Method of travel to work_Taxi"
## [126] "Method of travel to work_Motorcycle; scooter or moped"
## [127] "Method of travel to work_Driving a car or van"
## [128] "Method of travel to work_Passenger in a car or van"
## [129] "Method of travel to work_Bicycle"
## [130] "Method of travel to work_On foot"
## [131] "Method of travel to work_Other method of travel to work"
## [132] "NSSEC_All usual residents aged 16+"
## [133] "NSSEC_Higher managerial admin and professional"

```

```

## [134] "NSSEC_Lower managerial admin and professional"
## [135] "NSSEC_Intermediate"
## [136] "NSSEC_Small employers, own account workers"
## [137] "NSSEC_Lower supervisory and technical"
## [138] "NSSEC_Semi routine"
## [139] "NSSEC_Routine"
## [140] "NSSEC_Never worked and long term unemployed"
## [141] "NSSEC_Full time students"
## [142] "occupation_All usual residents aged 16 and over in employment"
## [143] "occupation_1. Managers, directors and senior officials"
## [144] "occupation_2. Professional occupations"
## [145] "occupation_3. Associate professional and technical occupations"
## [146] "occupation_4. Adminis-trative and secretarial occupations"
## [147] "occupation_5. Skilled trades occupations"
## [148] "occupation_6. Caring, leisure and other service occupations"
## [149] "occupation_7. Sales and customer service occupations"
## [150] "occupation_8. Process, plant and machine operatives"
## [151] "occupation_9. Elementary occupations"
## [152] "Age on arrival in UK_All usual residents"
## [153] "Age on arrival in UK_Born in the UK"
## [154] "Age on arrival in UK_Aged 0 to 4"
## [155] "Age on arrival in UK_Aged 5 to 7"
## [156] "Age on arrival in UK_Aged 8 to 9"
## [157] "Age on arrival in UK_Aged 10 to 14"
## [158] "Age on arrival in UK_Aged 15"
## [159] "Age on arrival in UK_Aged 16 to 17"
## [160] "Age on arrival in UK_Aged 18 to 19"
## [161] "Age on arrival in UK_Aged 20 to 24"
## [162] "Age on arrival in UK_Aged 25 to 29"
## [163] "Age on arrival in UK_Aged 30 to 44"
## [164] "Age on arrival in UK_Aged 45 to 59"
## [165] "Age on arrival in UK_Aged 60 to 64"
## [166] "Age on arrival in UK_Aged 65 to 74"
## [167] "Age on arrival in UK_Aged 75 to 84"
## [168] "Age on arrival in UK_Aged 85 to 89"
## [169] "Age on arrival in UK_Aged 90 or over"
## [170] "Country of birth_All Usual residents"
## [171] "Country of birth_United Kingdom"
## [172] "Country of birth_European Union EU14"
## [173] "Country of birth_Other EU countries"
## [174] "Country of birth_non-EU Europe"
## [175] "Country of birth_Africa"
## [176] "Country of birth_Middle East and Asia"
## [177] "Country of birth_Americas and Caribbean"
## [178] "Country of birth_Australasia, other Oceania and Other including British Overseas Territories"
## [179] "Five year age bands_All usual residents"
## [180] "Five year age bands_Aged 4 and under"
## [181] "Five year age bands_Aged 5 to 9"
## [182] "Five year age bands_Aged 10 to 14"
## [183] "Five year age bands_Aged 15 to 19"
## [184] "Five year age bands_Aged 20 to 24"
## [185] "Five year age bands_Aged 25 to 29"
## [186] "Five year age bands_Aged 30 to 34"
## [187] "Five year age bands_Aged 35 to 39"

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[188] "Five year age bands_Aged 40 to 44"
 ## [189] "Five year age bands_Aged 45 to 49"
 ## [190] "Five year age bands_Aged 50 to 54"
 ## [191] "Five year age bands_Aged 55 to 59"
 ## [192] "Five year age bands_Aged 60 to 64"
 ## [193] "Five year age bands_Aged 65 to 69"
 ## [194] "Five year age bands_Aged 70 to 74"
 ## [195] "Five year age bands_Aged 75 to 79"
 ## [196] "Five year age bands_Aged 80 to 84"
 ## [197] "Five year age bands_Aged 85 and over"
 ## [198] "Household composition_All households"
 ## [199] "Household composition_One person Aged 66+"
 ## [200] "Household composition_One person Aged up to 65"
 ## [201] "Household composition_Family: all aged 66+"
 ## [202] "Household composition_Married or civil partnership couple: No children"
 ## [203] "Household composition_Married or civil partnership couple: Dependent children"
 ## [204] "Household composition_Married or civil partnership couple: non-dependent children"
 ## [205] "Household composition_Cohabiting couple: No children"
 ## [206] "Household composition_Cohabiting couple: Dependent children"
 ## [207] "Household composition_Cohabiting couple: Non-dependent children"
 ## [208] "Household composition_Lone parent: dependent children"
 ## [209] "Household composition_Lone parent: non-dependent children"
 ## [210] "Household composition_Other with dependent children"
 ## [211] "Household composition_All other types"
 ## [212] "Household deprivation_All Households"
 ## [213] "Household deprivation_deprived in: no dimensions"
 ## [214] "Household deprivation_1 dimension"
 ## [215] "Household deprivation_2 dimensions"
 ## [216] "Household deprivation_3 dimensions"
 ## [217] "Household deprivation_4 dimensions"
 ## [218] "Household size_All Households"
 ## [219] "Household size_1 person in household"
 ## [220] "Household size_2 people in household"
 ## [221] "Household size_3 people in household"
 ## [222] "Household size_4 people in household"
 ## [223] "Household size_5 people in household"
 ## [224] "Household size_6 people in household"
 ## [225] "Household size_7 people in household"
 ## [226] "Household size_8 or more people in household"
 ## [227] "English Proficiency_Usual residents aged 3+"
 ## [228] "English Proficiency_Main language is English"
 ## [229] "English Proficiency_Language not English: Speak English very well"
 ## [230] "English Proficiency_Language not English: Speak English well"
 ## [231] "English Proficiency_Language not English: Cannot speak English well"
 ## [232] "English Proficiency_Language not English: Cannot speak English"
 ## [233] "Ethnic group_All usual residents"
 ## [234] "Ethnic group_White British"
 ## [235] "Ethnic group_White Irish"
 ## [236] "Ethnic group_White Gypsy/Irish Traveller"
 ## [237] "Ethnic group_White Roma"
 ## [238] "Ethnic group_White Other"
 ## [239] "Ethnic group_Mixed White and Asian"
 ## [240] "Ethnic group_Mixed White and Black African"
 ## [241] "Ethnic group_Mixed White and Black Caribbean"

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## [242] "Ethnic group_Mixed Other"
## [243] "Ethnic group_Asian Bangladeshi"
## [244] "Ethnic group_Asian Chinese"
## [245] "Ethnic group_Asian Indian"
## [246] "Ethnic group_Asian Pakistani"
## [247] "Ethnic group_Asian Other"
## [248] "Ethnic group_Black African"
## [249] "Ethnic group_Black Caribbean"
## [250] "Ethnic group_Black Other"
## [251] "Ethnic group_Other Arab"
## [252] "Ethnic group_Other Any other"
## [253] "Household Language_All households"
## [254] "Household Language_All people 16+ have English as main language"
## [255] "Household Language_At least one but not all people 16+ have English as main language"
## [256] "Household Language_No people 16+ but one or more aged 3-15 has English as main language"
## [257] "Household Language_No people have English as main language"
## [258] "Multiple ethnic groups_All households"
## [259] "Multiple ethnic groups_One person household"
## [260] "Multiple ethnic groups_Same ethnic group"
## [261] "Multiple ethnic groups_Ethnic groups differ across generations only"
## [262] "Multiple ethnic groups_Ethnic groups differ within partnerships"
## [263] "Multiple ethnic groups_Other multiple ethnic groups"
## [264] "Multiple religions in household_All households"
## [265] "Multiple religions in household_One-person household"
## [266] "Multiple religions in household_No people answered the question"
## [267] "Multiple religions in household_No religion (may include religion not stated)"
## [268] "Multiple religions in household_One religion only (may include religion not stated)"
## [269] "Multiple religions in household_One religion and no religion (may include religion not stated)"
## [270] "Multiple religions in household_Two or more religions (may include no religion and religion)"
## [271] "accommodation type_All households"
## [272] "accommodation type_Detached"
## [273] "accommodation type_Semi-detached"
## [274] "accommodation type_Terraced"
## [275] "accommodation type_Purpose built flat"
## [276] "accommodation type_Flat in a converted/ shared house (includes all households in shared dwell)"
## [277] "accommodation type_Flat in a commercial building"
## [278] "accommodation type_Caravan / other mobile or temporary structure"
## [279] "cars or vans_All households"
## [280] "cars or vans_none"
## [281] "cars or vans_one"
## [282] "cars or vans_two"
## [283] "cars or vans_three or more"
## [284] "central heating_All households"
## [285] "central heating_No central heating"
## [286] "central heating_Gas"
## [287] "central heating_Electric"
## [288] "central heating_Oil"
## [289] "central heating_Solid fuel"
## [290] "central heating_Other including communal and renewable"
## [291] "central heating_Two or more types"
## [292] "number of bedrooms_All households"
## [293] "number of bedrooms_1 bedroom"
## [294] "number of bedrooms_2 bedrooms"
## [295] "number of bedrooms_3 bedrooms"

```

```

## [296] "number of bedrooms_4 or more bedrooms"
## [297] "number of rooms_All households"
## [298] "number of rooms_1 room"
## [299] "number of rooms_2 rooms"
## [300] "number of rooms_3 rooms"
## [301] "number of rooms_4 rooms"
## [302] "number of rooms_5 rooms"
## [303] "number of rooms_6 rooms"
## [304] "number of rooms_7 rooms"
## [305] "number of rooms_8 rooms"
## [306] "number of rooms_9 or more rooms"
## [307] "occupancy rating - bedrooms_All Households"
## [308] "occupancy rating - bedrooms_Occupancy rating: +2 or more"
## [309] "occupancy rating - bedrooms_+1"
## [310] "occupancy rating - bedrooms_0"
## [311] "occupancy rating - bedrooms_-1"
## [312] "occupancy rating - bedrooms_-2 or less"
## [313] "tenure - households_All Households"
## [314] "tenure - households_Owned outright"
## [315] "tenure - households_Owned with a mortgage or loan"
## [316] "tenure - households_Shared ownership"
## [317] "tenure - households_Rented from Local Authority"
## [318] "tenure - households_Other social rented"
## [319] "tenure - households_Private landlord or letting agency"
## [320] "tenure - households_Other private rented"
## [321] "tenure - households_Rent free"
## [322] "Disability_All Usual residents"
## [323] "Disability_Disabled; activities limited a lot"
## [324] "Disability_Disabled; activities limited a little"
## [325] "Disability_Not disabled"
## [326] "General health_All Usual residents"
## [327] "General health_Very good health"
## [328] "General health_Good health"
## [329] "General health_Fair health"
## [330] "General health_Bad health"
## [331] "General health_Very bad health"
## [332] "Qualifications_Usual residents aged 16+"
## [333] "Qualifications_none"
## [334] "Qualifications_Level 1"
## [335] "Qualifications_Level 2"
## [336] "Qualifications_Apprentice-ship"
## [337] "Qualifications_Level 3"
## [338] "Qualifications_Level 4+"
## [339] "Qualifications_Other"
## [340] "Unpaid care_All Usual residents aged 5+"
## [341] "Unpaid care_Provides no care"
## [342] "Unpaid care_1 to 19 hours"
## [343] "Unpaid care_20 to 49 hours"
## [344] "Unpaid care_50 or more hours"

```

YEAR 2011 DATA ALL MERGED.

```
head(london_2011_supra)
```

```
## # A tibble: 6 x 343
```

```
##   fullAddress      postcode country outcode latitude longitude bathrooms bedrooms
##   <chr>            <chr>      <chr>  <chr>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 Flat 9, 35 Fur~ EC4A 1JQ England EC4A      51.5    -0.110          1          1
## 2 Flat 6, 35 Fur~ EC4A 1JQ England EC4A      51.5    -0.110          NA          NA
## 3 Flat 35, Octav~ SW1P 2TA England SW1P      51.5    -0.132          2          2
## 4 24 Chester Cou~ SE5 7HS  England SE5       51.5    -0.0922         1          1
## 5 18 Alexandra G~ N10 3RL  England N10      51.6    -0.140          1          4
## 6 17 Collins Str~ SE3 OUG  England SE3      51.5     0.00697         1          2
## # i 335 more variables: floorAreaSqM <dbl>, livingRooms <dbl>, tenure <chr>,
## #   propertyType <chr>, currentEnergyRating <chr>,
## #   rentEstimate_lowerPrice <dbl>, rentEstimate_currentPrice <dbl>,
## #   rentEstimate_upperPrice <dbl>, saleEstimate_lowerPrice <dbl>,
## #   saleEstimate_currentPrice <dbl>, saleEstimate_upperPrice <dbl>,
## #   saleEstimate_confidenceLevel <chr>, saleEstimate_ingestedAt <dtm>,
## #   saleEstimate_valueChange.numericChange <dbl>, ...
```

```
colnames(london_2011_supra)
```

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##   [1] "fullAddress"
##   [2] "postcode"
##   [3] "country"
##   [4] "outcode"
##   [5] "latitude"
##   [6] "longitude"
##   [7] "bathrooms"
##   [8] "bedrooms"
##   [9] "floorAreaSqM"
##  [10] "livingRooms"
##  [11] "tenure"
##  [12] "propertyType"
##  [13] "currentEnergyRating"
##  [14] "rentEstimate_lowerPrice"
##  [15] "rentEstimate_currentPrice"
##  [16] "rentEstimate_upperPrice"
##  [17] "saleEstimate_lowerPrice"
##  [18] "saleEstimate_currentPrice"
##  [19] "saleEstimate_upperPrice"
##  [20] "saleEstimate_confidenceLevel"
##  [21] "saleEstimate_ingestedAt"
##  [22] "saleEstimate_valueChange.numericChange"
##  [23] "saleEstimate_valueChange.percentageChange"
##  [24] "saleEstimate_valueChange.saleDate"
##  [25] "history_date"
##  [26] "history_price"
##  [27] "history_percentageChange"
##  [28] "history_numericChange"
##  [29] "In Use?"
##  [30] "Latitude"
##  [31] "Longitude"
##  [32] "Easting"
##  [33] "Northing"
##  [34] "Grid Ref"
##  [35] "County"
##  [36] "District"
##  [37] "Ward"
```

```

## [38] "District Code"
## [39] "Ward Code"
## [40] "Country"
## [41] "County Code"
## [42] "Introduced"
## [43] "Terminated"
## [44] "Parish"
## [45] "National Park"
## [46] "Population"
## [47] "Households"
## [48] "Built up area"
## [49] "Lower layer super output area"
## [50] "Rural/urban"
## [51] "Region"
## [52] "Altitude"
## [53] "London zone"
## [54] "LSOA Code"
## [55] "Local authority"
## [56] "MSOA Code"
## [57] "Middle layer super output area"
## [58] "Parish Code"
## [59] "Census output area"
## [60] "Index of Multiple Deprivation"
## [61] "Quality"
## [62] "User Type"
## [63] "Last updated"
## [64] "Nearest station"
## [65] "Distance to station"
## [66] "Postcode area"
## [67] "Postcode district"
## [68] "Police force"
## [69] "Water company"
## [70] "Plus Code"
## [71] "Average Income"
## [72] "Sewage Company"
## [73] "Travel To Work Area"
## [74] "ITL level 2"
## [75] "ITL level 3"
## [76] "UPRNs"
## [77] "Distance to sea"
## [78] "LSOA21 Code"
## [79] "Lower layer super output area 2021"
## [80] "MSOA21 Code"
## [81] "Middle layer super output area 2021"
## [82] "Census output area 2021"
## [83] "IMD decile"
## [84] "Constituency Code 2024"
## [85] "Constituency Name 2024"
## [86] "Property Type"
## [87] "Roads"
## [88] "local authority code"
## [89] "local authority name"
## [90] "Distance traveled to work_All usual residents aged 16-74 in employment"
## [91] "Distance traveled to work_Less than 2km"

```

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## [92] "Distance traveled to work_2km to less than 5km"
## [93] "Distance traveled to work_5km to less than 10km"
## [94] "Distance traveled to work_10km to less than 20km"
## [95] "Distance traveled to work_20km to less than 30km"
## [96] "Distance traveled to work_30km to less than 40km"
## [97] "Distance traveled to work_40km to less than 60km"
## [98] "Distance traveled to work_60km and over"
## [99] "Distance traveled to work_Work mainly at or from home"
## [100] "Distance traveled to work_Other"
## [101] "Economic Activity_All usual residents aged 16-74"
## [102] "Economic Activity_Economically active: Employee: Full-time"
## [103] "Economic Activity_Economically active: Employee: Part-time"
## [104] "Economic Activity_Economically active: Full-time student"
## [105] "Economic Activity_Economically active: Self-employed with employees: Full-time"
## [106] "Economic Activity_Economically active: Self-employed with employees: Part-time"
## [107] "Economic Activity_Economically active: Self-employed without employees: Full-time"
## [108] "Economic Activity_Economically active: Self-employed without employees: Part-time"
## [109] "Economic Activity_Economically active: Unemployed"
## [110] "Economic Activity_Economically inactive: Long-term sick or disabled"
## [111] "Economic Activity_Economically inactive: Looking after home or family"
## [112] "Economic Activity_Economically inactive: Other"
## [113] "Economic Activity_Economically inactive: Retired"
## [114] "Economic Activity_Economically inactive: Full-time students"
## [115] "hours worked_All usual residents aged 16-74 in employment"
## [116] "hours worked_15 hours or less"
## [117] "hours worked_16 to 30 hours"
## [118] "hours worked_31 to 48 hours"
## [119] "hours worked_49 or more hours"
## [120] "Method of travel to work_All usual residents aged 16-74 in work"
## [121] "Method of travel to work_Work mainly at or from home"
## [122] "Method of travel to work_Underground; metro; light rail; tram"
## [123] "Method of travel to work_Train"
## [124] "Method of travel to work_Bus; minibus or coach"
## [125] "Method of travel to work_Taxi"
## [126] "Method of travel to work_Motorcycle; scooter or moped"
## [127] "Method of travel to work_Driving a car or van"
## [128] "Method of travel to work_Passenger in a car or van"
## [129] "Method of travel to work_Bicycle"
## [130] "Method of travel to work_On foot"
## [131] "Method of travel to work_Other method of travel to work"
## [132] "NSSEC_All usual residents aged 16-74"
## [133] "NSSEC_Higher managerial admin and professional"
## [134] "NSSEC_Lower managerial admin and professional"
## [135] "NSSEC_Intermediate"
## [136] "NSSEC_Small employers, own account workers"
## [137] "NSSEC_Lower supervisory and technical"
## [138] "NSSEC_Semi routine"
## [139] "NSSEC_Routine"
## [140] "NSSEC_Never worked and long term unemployed"
## [141] "NSSEC_Full time students"
## [142] "occupation_All usual residents aged 16-74 in employment"
## [143] "occupation_1. Managers, directors and senior officials"
## [144] "occupation_2. Professional occupations"
## [145] "occupation_3. Associate professional and technical occupations"

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## [146] "occupation_4. Adminis-trative and secretarial occupations"
## [147] "occupation_5. Skilled trades occupations"
## [148] "occupation_6. Caring, leisure and other service occupations"
## [149] "occupation_7. Sales and customer service occupations"
## [150] "occupation_8. Process, plant and machine operatives"
## [151] "occupation_9. Elementary occupations"
## [152] "Age on arrival in UK_All usual residents"
## [153] "Age on arrival in UK_Born in the UK"
## [154] "Age on arrival in UK_Aged 0 to 4"
## [155] "Age on arrival in UK_Aged 5 to 7"
## [156] "Age on arrival in UK_Aged 8 to 9"
## [157] "Age on arrival in UK_Aged 10 to 14"
## [158] "Age on arrival in UK_Aged 15"
## [159] "Age on arrival in UK_Aged 16 to 17"
## [160] "Age on arrival in UK_Aged 18 to 19"
## [161] "Age on arrival in UK_Aged 20 to 24"
## [162] "Age on arrival in UK_Aged 25 to 29"
## [163] "Age on arrival in UK_Aged 30 to 44"
## [164] "Age on arrival in UK_Aged 45 to 59"
## [165] "Age on arrival in UK_Aged 60 to 64"
## [166] "Age on arrival in UK_Aged 65 to 74"
## [167] "Age on arrival in UK_Aged 75 to 84"
## [168] "Age on arrival in UK_Aged 85 to 89"
## [169] "Age on arrival in UK_Aged 90 or over"
## [170] "Country of birth_All Usual residents"
## [171] "Country of birth_United Kingdom"
## [172] "Country of birth_European Union EU14"
## [173] "Country of birth_Other EU countries"
## [174] "Country of birth_non-EU Europe"
## [175] "Country of birth_Africa"
## [176] "Country of birth_Middle East and Asia"
## [177] "Country of birth_Americas and Caribbean"
## [178] "Country of birth_Australasia, other Oceania and Other including British Overseas Territories"
## [179] "Five year age bands_All usual residents"
## [180] "Five year age bands_Aged 4 and under"
## [181] "Five year age bands_Aged 5 to 9"
## [182] "Five year age bands_Aged 10 to 14"
## [183] "Five year age bands_Aged 15 to 19"
## [184] "Five year age bands_Aged 20 to 24"
## [185] "Five year age bands_Aged 25 to 29"
## [186] "Five year age bands_Aged 30 to 34"
## [187] "Five year age bands_Aged 35 to 39"
## [188] "Five year age bands_Aged 40 to 44"
## [189] "Five year age bands_Aged 45 to 49"
## [190] "Five year age bands_Aged 50 to 54"
## [191] "Five year age bands_Aged 55 to 59"
## [192] "Five year age bands_Aged 60 to 64"
## [193] "Five year age bands_Aged 65 to 69"
## [194] "Five year age bands_Aged 70 to 74"
## [195] "Five year age bands_Aged 75 to 79"
## [196] "Five year age bands_Aged 80 to 84"
## [197] "Five year age bands_Aged 85 and over"
## [198] "Household composition_All households"
## [199] "Household composition_One person Aged 65+"

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[200] "Household composition_One person Aged up to 64"
 ## [201] "Household composition_Family: all aged 65+"
 ## [202] "Household composition_Married or civil partnership couple: No children"
 ## [203] "Household composition_Married or civil partnership couple: Dependent children"
 ## [204] "Household composition_Married or civil partnership couple: non-dependent children"
 ## [205] "Household composition_Cohabiting couple: No children"
 ## [206] "Household composition_Cohabiting couple: Dependent children"
 ## [207] "Household composition_Cohabiting couple: Non-dependent children"
 ## [208] "Household composition_Lone parent: dependent children"
 ## [209] "Household composition_Lone parent: non-dependent children"
 ## [210] "Household composition_Other with dependent children"
 ## [211] "Household composition_All other types"
 ## [212] "Household deprivation_All Households"
 ## [213] "Household deprivation_deprived in: no dimensions"
 ## [214] "Household deprivation_1 dimension"
 ## [215] "Household deprivation_2 dimensions"
 ## [216] "Household deprivation_3 dimensions"
 ## [217] "Household deprivation_4 dimensions"
 ## [218] "Household size_All Households"
 ## [219] "Household size_1 person in household"
 ## [220] "Household size_2 people in household"
 ## [221] "Household size_3 people in household"
 ## [222] "Household size_4 people in household"
 ## [223] "Household size_5 people in household"
 ## [224] "Household size_6 people in household"
 ## [225] "Household size_7 people in household"
 ## [226] "Household size_8 or more people in household"
 ## [227] "English Proficiency_Usual residents aged 3+"
 ## [228] "English Proficiency_Main language is English"
 ## [229] "English Proficiency_Language not English: Speak English very well"
 ## [230] "English Proficiency_Language not English: Speak English well"
 ## [231] "English Proficiency_Language not English: Cannot speak English well"
 ## [232] "English Proficiency_Language not English: Cannot speak English"
 ## [233] "Ethnic group_All usual residents"
 ## [234] "Ethnic group_White British"
 ## [235] "Ethnic group_White Irish"
 ## [236] "Ethnic group_White Gypsy/Irish Traveller"
 ## [237] "Ethnic group_...8"
 ## [238] "Ethnic group_White Other"
 ## [239] "Ethnic group_Mixed White and Asian"
 ## [240] "Ethnic group_Mixed White and Black African"
 ## [241] "Ethnic group_Mixed White and Black Caribbean"
 ## [242] "Ethnic group_Mixed Other"
 ## [243] "Ethnic group_Asian Bangladeshi"
 ## [244] "Ethnic group_Asian Chinese"
 ## [245] "Ethnic group_Asian Indian"
 ## [246] "Ethnic group_Asian Pakistani"
 ## [247] "Ethnic group_Asian Other"
 ## [248] "Ethnic group_Black African"
 ## [249] "Ethnic group_Black Caribbean"
 ## [250] "Ethnic group_Black Other"
 ## [251] "Ethnic group_Other Arab"
 ## [252] "Ethnic group_Other Any other"
 ## [253] "Household Language_All households"

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## [254] "Household Language_All people 16+ have English as main language"
## [255] "Household Language_At least one but not all people 16+ have English as main language"
## [256] "Household Language_No people 16+ but one or more aged 3-15 has English as main language"
## [257] "Household Language_No people have English as main language"
## [258] "Multiple ethnic groups_All households.x"
## [259] "Multiple ethnic groups_One person household.x"
## [260] "Multiple ethnic groups_Same ethnic group.x"
## [261] "Multiple ethnic groups_Ethnic groups differ across generations only.x"
## [262] "Multiple ethnic groups_Ethnic groups differ within partnerships.x"
## [263] "Multiple ethnic groups_Other multiple ethnic groups.x"
## [264] "Multiple ethnic groups_All households.y"
## [265] "Multiple ethnic groups_One person household.y"
## [266] "Multiple ethnic groups_Same ethnic group.y"
## [267] "Multiple ethnic groups_Ethnic groups differ across generations only.y"
## [268] "Multiple ethnic groups_Ethnic groups differ within partnerships.y"
## [269] "Multiple ethnic groups_Other multiple ethnic groups.y"
## [270] "accommodation type_All households"
## [271] "accommodation type_Detached"
## [272] "accommodation type_Semi-detached"
## [273] "accommodation type_Terraced"
## [274] "accommodation type_Purpose built flat"
## [275] "accommodation type_Flat in a converted/ shared house (includes all households in shared dwell."
## [276] "accommodation type_Flat in a commercial building"
## [277] "accommodation type_Caravan / other mobile or temporary structure"
## [278] "cars or vans_All households"
## [279] "cars or vans_none"
## [280] "cars or vans_one"
## [281] "cars or vans_two"
## [282] "cars or vans_three or more"
## [283] "central heating_All households"
## [284] "central heating_No central heating"
## [285] "central heating_Gas"
## [286] "central heating_Electric"
## [287] "central heating_Oil"
## [288] "central heating_Solid fuel"
## [289] "central heating_Other including communal and renewable"
## [290] "central heating_Two or more types"
## [291] "number of bedrooms_All households"
## [292] "number of bedrooms_1 bedroom"
## [293] "number of bedrooms_2 bedrooms"
## [294] "number of bedrooms_3 bedrooms"
## [295] "number of bedrooms_4 or more bedrooms"
## [296] "number of rooms_All households"
## [297] "number of rooms_1 room"
## [298] "number of rooms_2 rooms"
## [299] "number of rooms_3 rooms"
## [300] "number of rooms_4 rooms"
## [301] "number of rooms_5 rooms"
## [302] "number of rooms_6 rooms"
## [303] "number of rooms_7 rooms"
## [304] "number of rooms_8 rooms"
## [305] "number of rooms_9 or more rooms"
## [306] "occupancy rating - bedrooms_All Households"
## [307] "occupancy rating - bedrooms_Occupancy rating: +2 or more"

```

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## [308] "occupancy rating - bedrooms_+1"
## [309] "occupancy rating - bedrooms_0"
## [310] "occupancy rating - bedrooms_-1"
## [311] "occupancy rating - bedrooms_-2 or less"
## [312] "tenure - households_All Households"
## [313] "tenure - households_Owned outright"
## [314] "tenure - households_Owned with a mortgage or loan"
## [315] "tenure - households_Shared ownership"
## [316] "tenure - households_Rented from Local Authority"
## [317] "tenure - households_Other social rented"
## [318] "tenure - households_Private landlord or letting agency"
## [319] "tenure - households_Other private rented"
## [320] "tenure - households_Rent free"
## [321] "Disability_All Usual residents"
## [322] "Disability_Disabled; activities limited a lot"
## [323] "Disability_Disabled; activities limited a little"
## [324] "Disability_Not disabled"
## [325] "General health_All Usual residents"
## [326] "General health_Very good health"
## [327] "General health_Good health"
## [328] "General health_Fair health"
## [329] "General health_Bad health"
## [330] "General health_Very bad health"
## [331] "Qualifications_Usual residents aged 16+"
## [332] "Qualifications_none"
## [333] "Qualifications_Level 1"
## [334] "Qualifications_Level 2"
## [335] "Qualifications_Apprentice-ship"
## [336] "Qualifications_Level 3"
## [337] "Qualifications_Level 4+"
## [338] "Qualifications_Other"
## [339] "Unpaid care_All Usual residents aged 5+"
## [340] "Unpaid care_Provides no care"
## [341] "Unpaid care_1 to 19 hours"
## [342] "Unpaid care_20 to 49 hours"
## [343] "Unpaid care_50 or more hours"

```

Some visualisations...

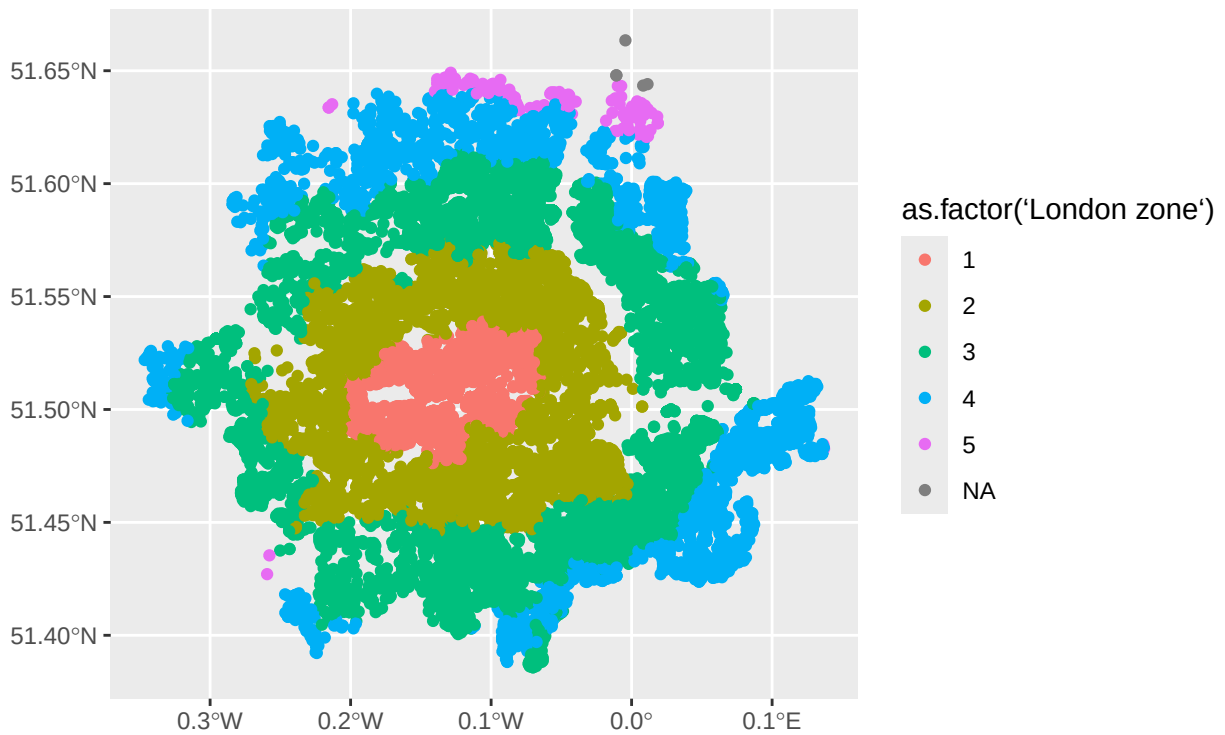


Figure 1: London Zone Map

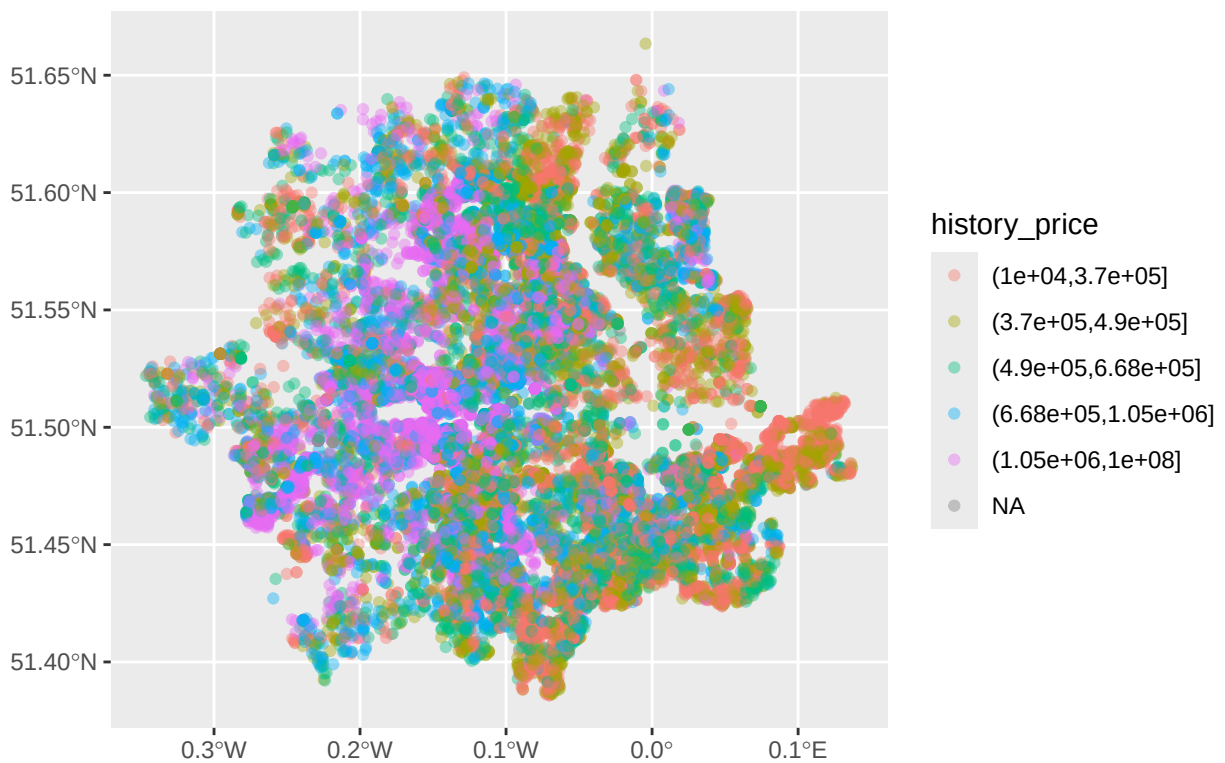


Figure 2: 2021 Property Prices

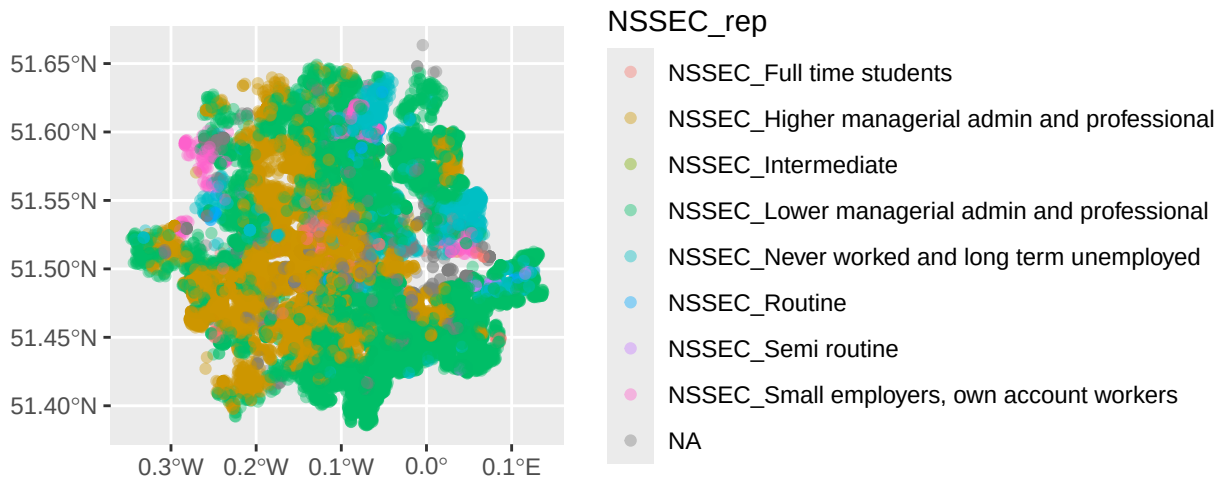


Figure 3: NSSEC by LSOA Code

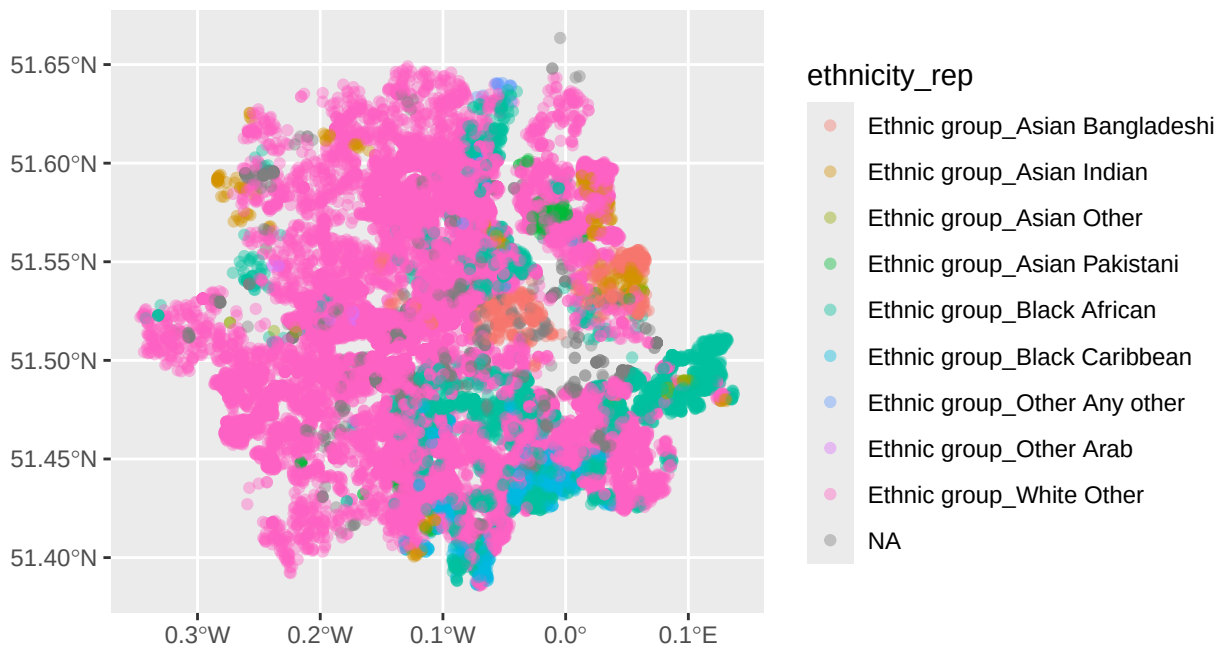


Figure 4: Biggest Ethnic Minority by LSOA Code